

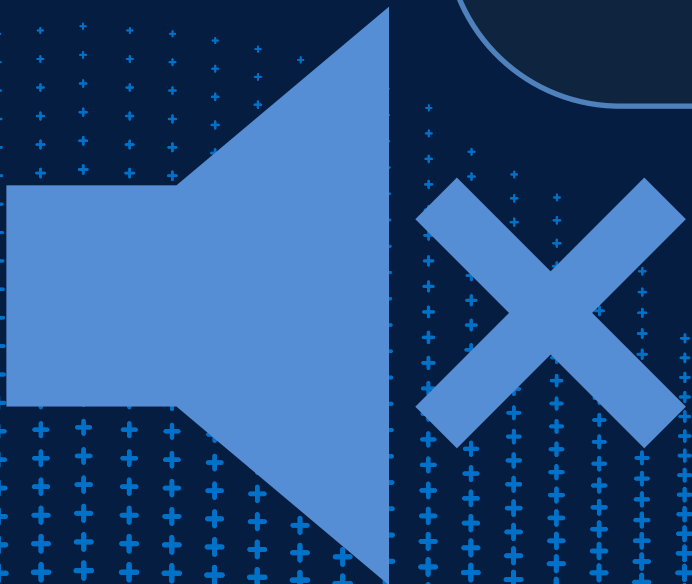
EPA's National Sewershed Dataset

January 14th, 2026



Housekeeping

- Please enter questions into the chat
- Mics have been disabled for this meeting
- This meeting will be recorded



Agenda



What is the sewershed dataset?



Use Cases



Methodology



Feedback Portal



Ongoing Development



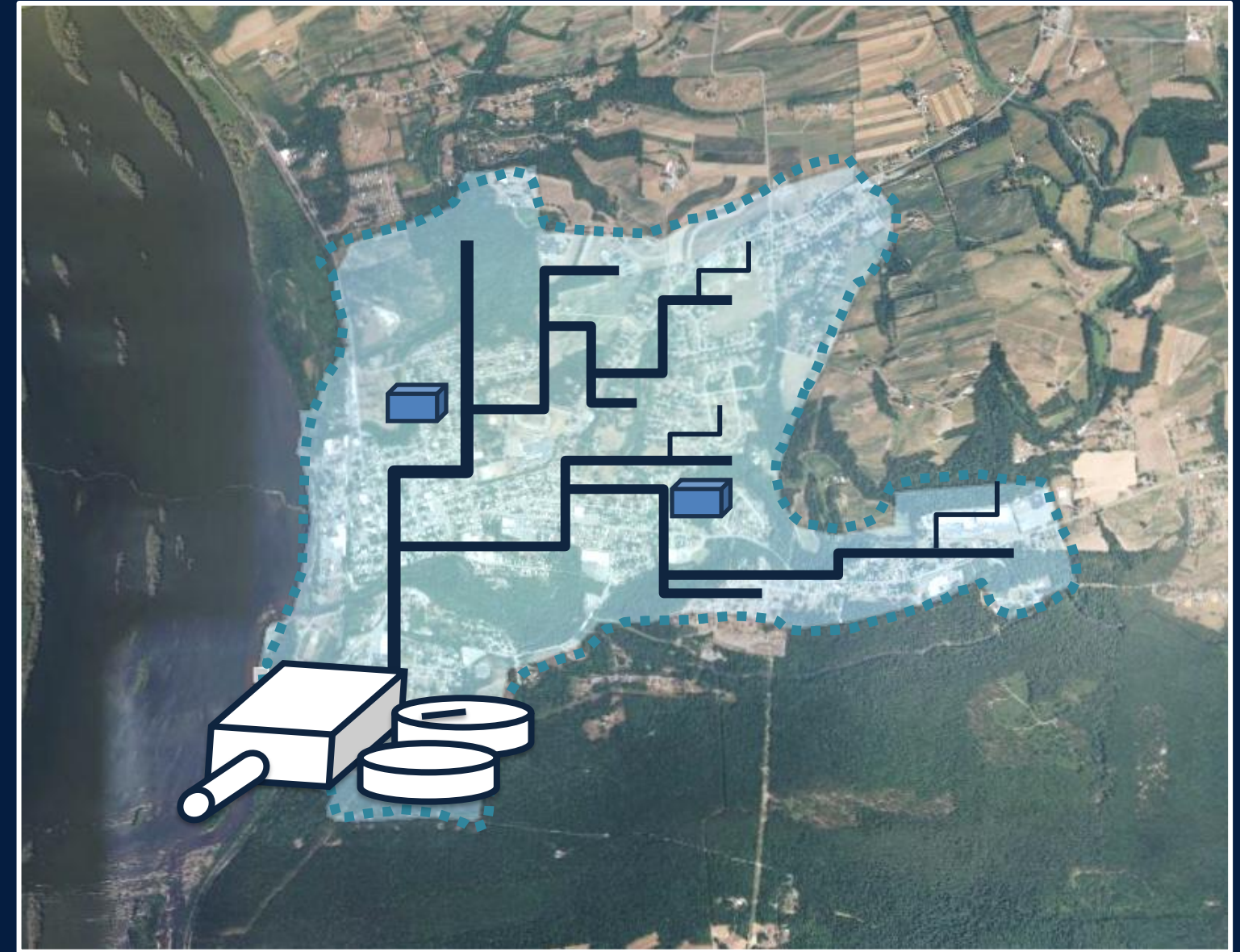
Demo of Web Map and Feedback Portal



Questions

What is it?

- Polygon shapefiles of POTWs' **sewersheds** – the geographic area where all the sewers flow to a single end wastewater treatment plant.
- EPA will iteratively update the dataset with public feedback and the results of future Clean Watershed Needs Surveys.



Which POTWs?

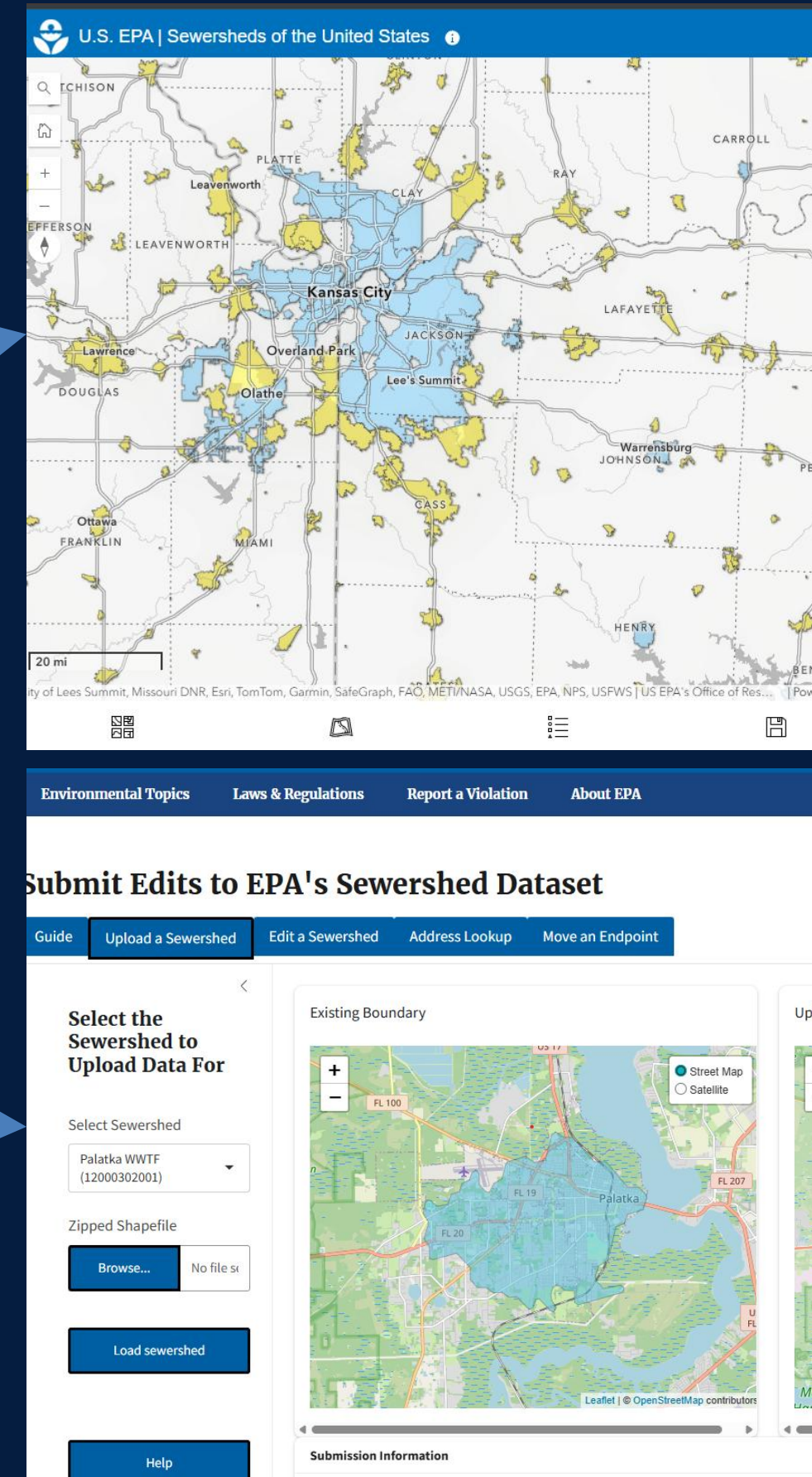
Endpoint facilities - the last point of treatment before discharging to the environment.

POTWs where less than 50% of their wastewater is sent to another facility.

What is it?

Products:

- **Online map application** that includes polygons for ~17,000 POTWs' sewersheds.
 - 19% are verified sewersheds – sourced from online and state partners
- **Feedback portal** where the public can make suggestions or submit data for consideration in future updates.



Use Cases

1

Utilities: provides information for targeting investments and encouraging smart development of community infrastructure - whether it is expanding distribution, consolidation, or **regionalization of POTWs**.

2

Decentralized (septic): identifies where gaps are and where to focus resources on **infrastructure funding**.

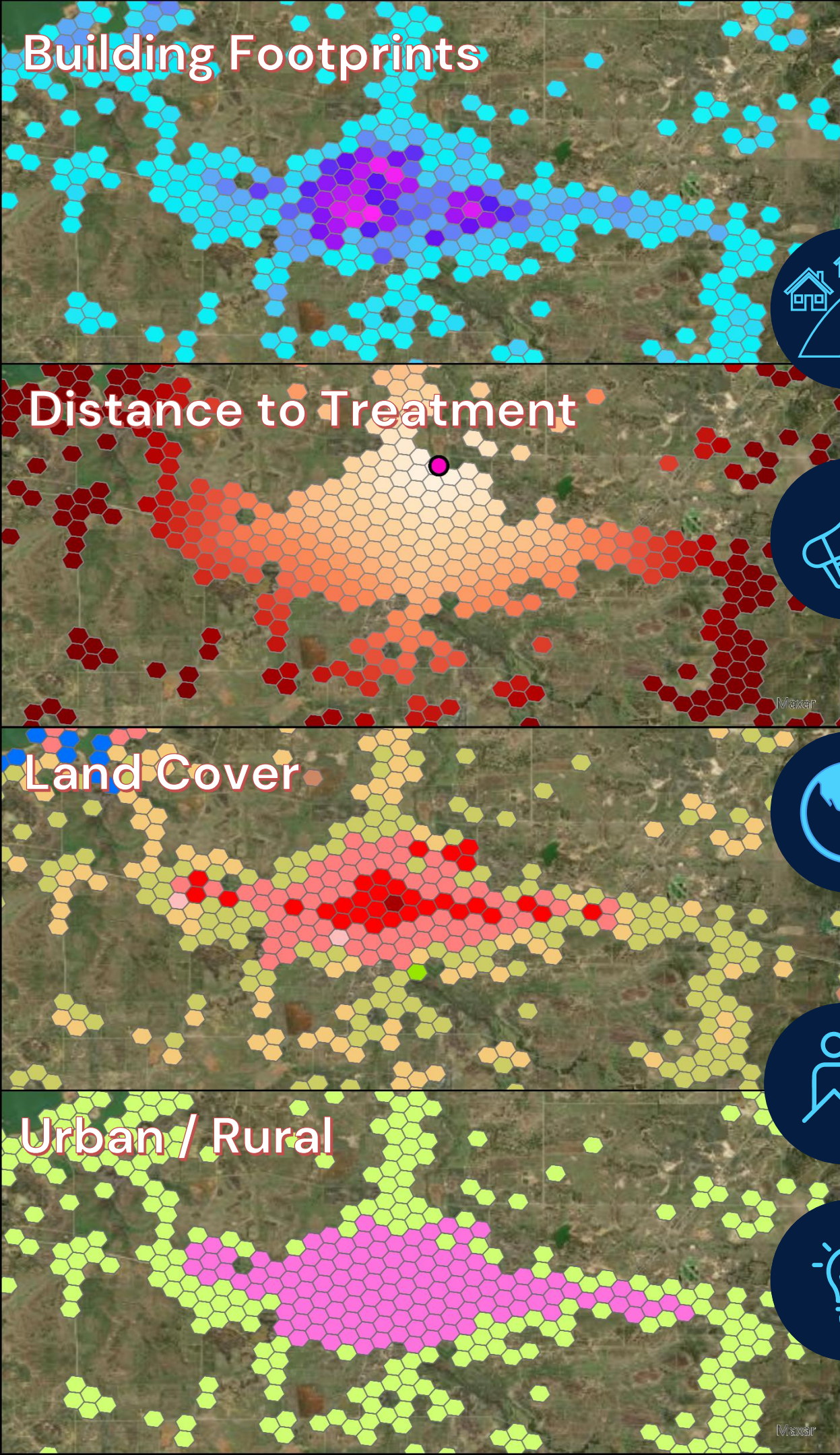
3

Wastewater Monitoring: identifies utilities with information on extent of their coverages and loads that may impact treatment plants

4

Emergency Response: Quickly targets assistance and identifies priority investments to speed up recovery.

Use Case: During natural disasters like Hurricane Helene, there is a need to quickly identify which people are affected by damaged wastewater infrastructure. The sewershed map can identify which communities are impacted for a given POTW to target resource deployment.



Methodology – Data Inputs



Physical Characteristics

Microsoft Building Footprints
% Impervious Surface



U.S. Census data

1990 Census – Last time we asked if people had sewer connection / Population / Housing Units

2020 Census – Population / Housing Units

Model derives 30-year changes in population and housing.



Environmental Variables

Land Cover
Elevation



Clean Watersheds Needs Survey

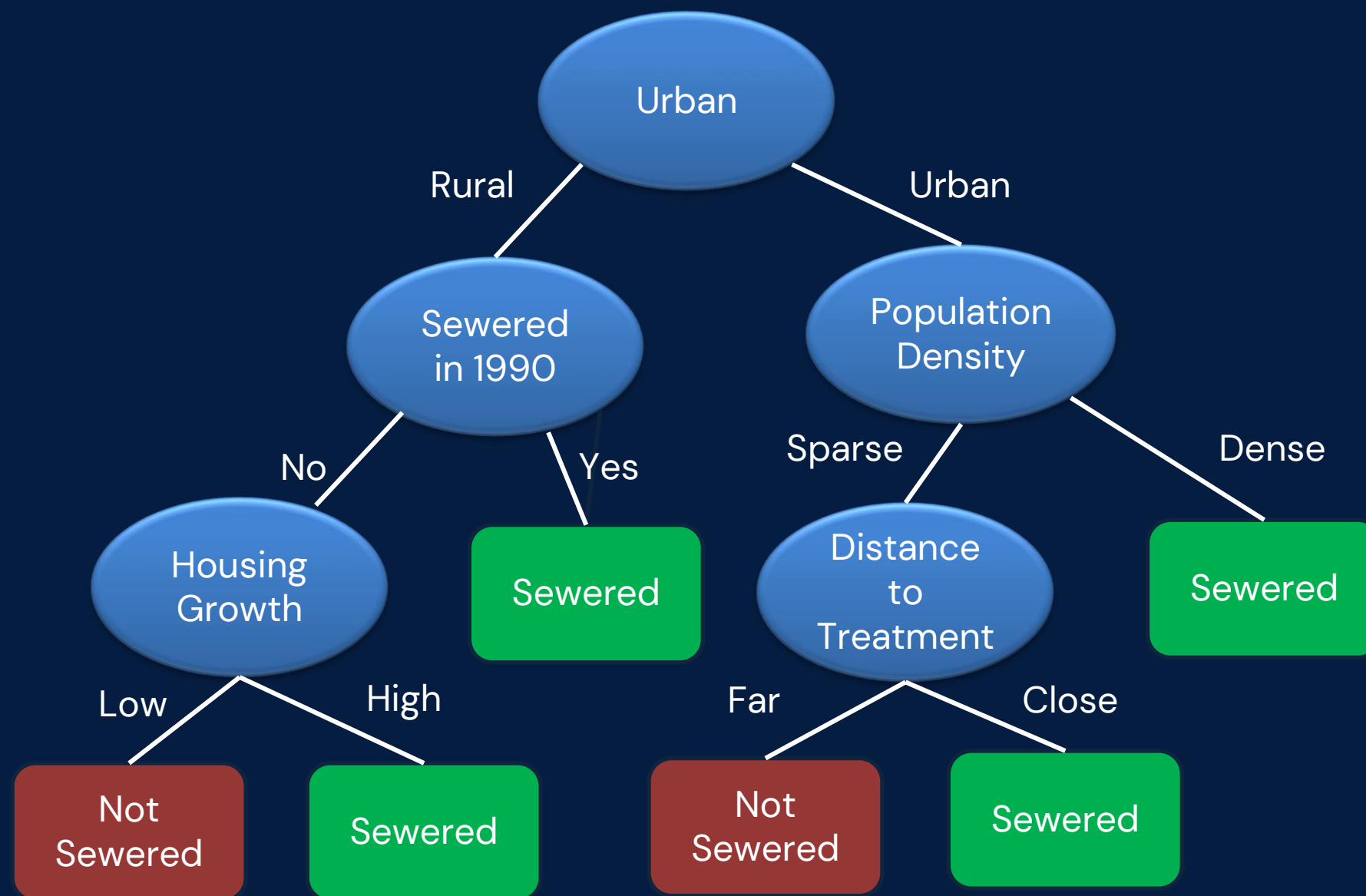
POTW locations, discharge type, flow, population served



Online and state-provided sewershed polygons

Model Development

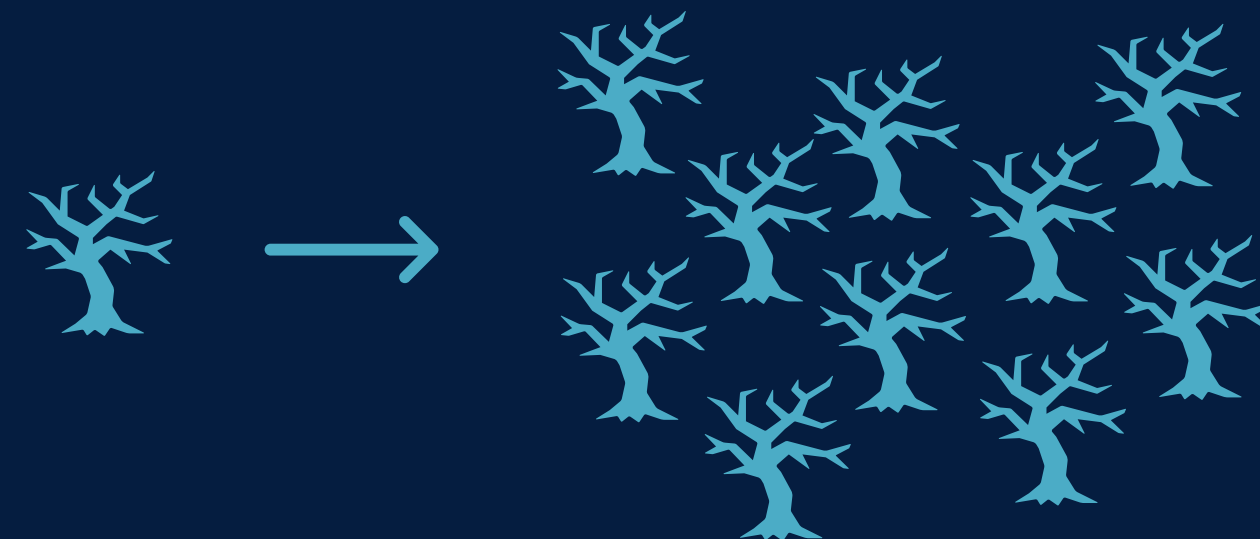
Goal: Predict the probability of a hexagon being sewered and being served by a specific treatment plant



Simplified Decision Tree for Illustration Purposes

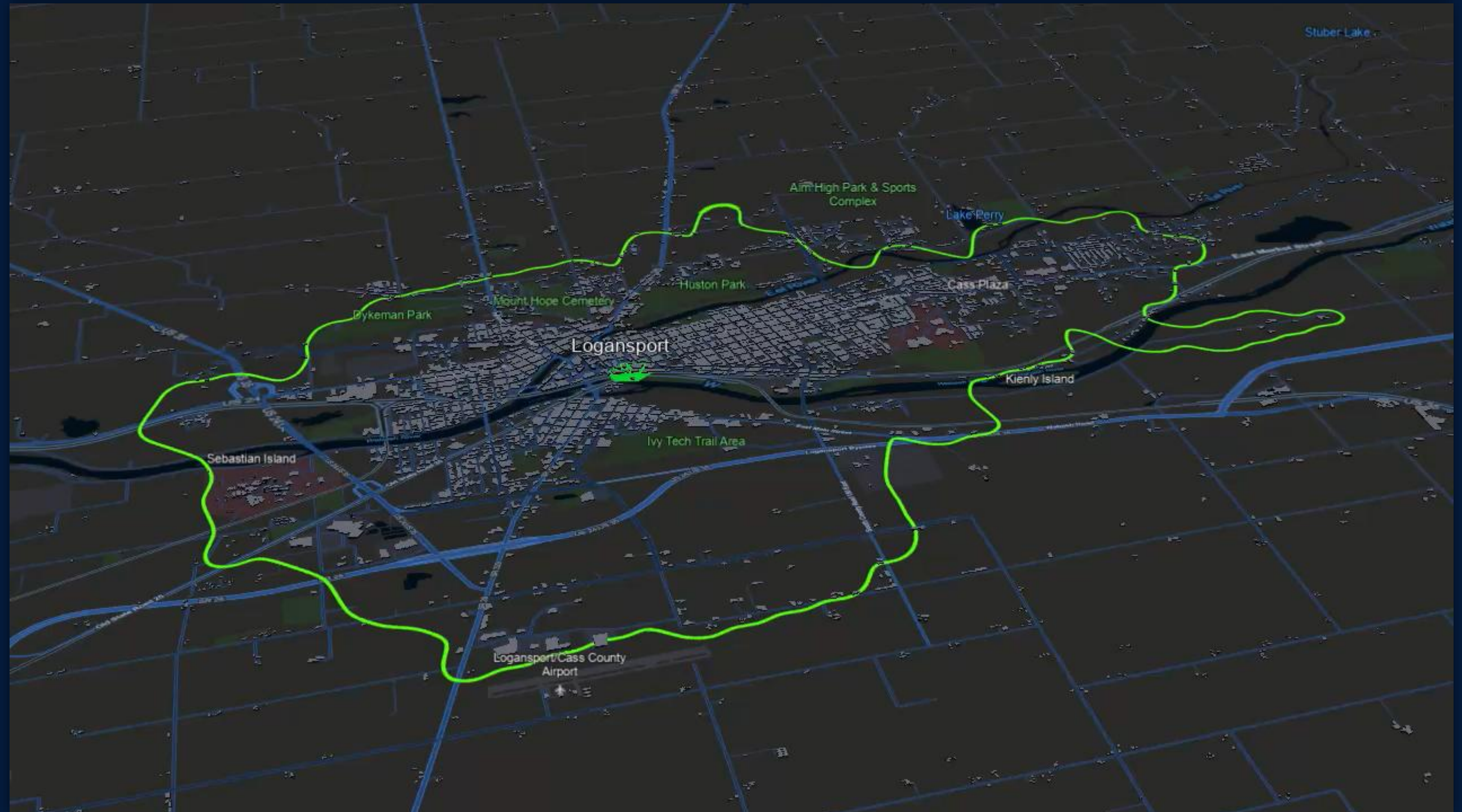
Boosted Decision Trees

- Starts with a single decision tree
- Creates additional trees, learning from its mistakes and successes after each new tree
- Continues building new trees until accuracy stops increasing.



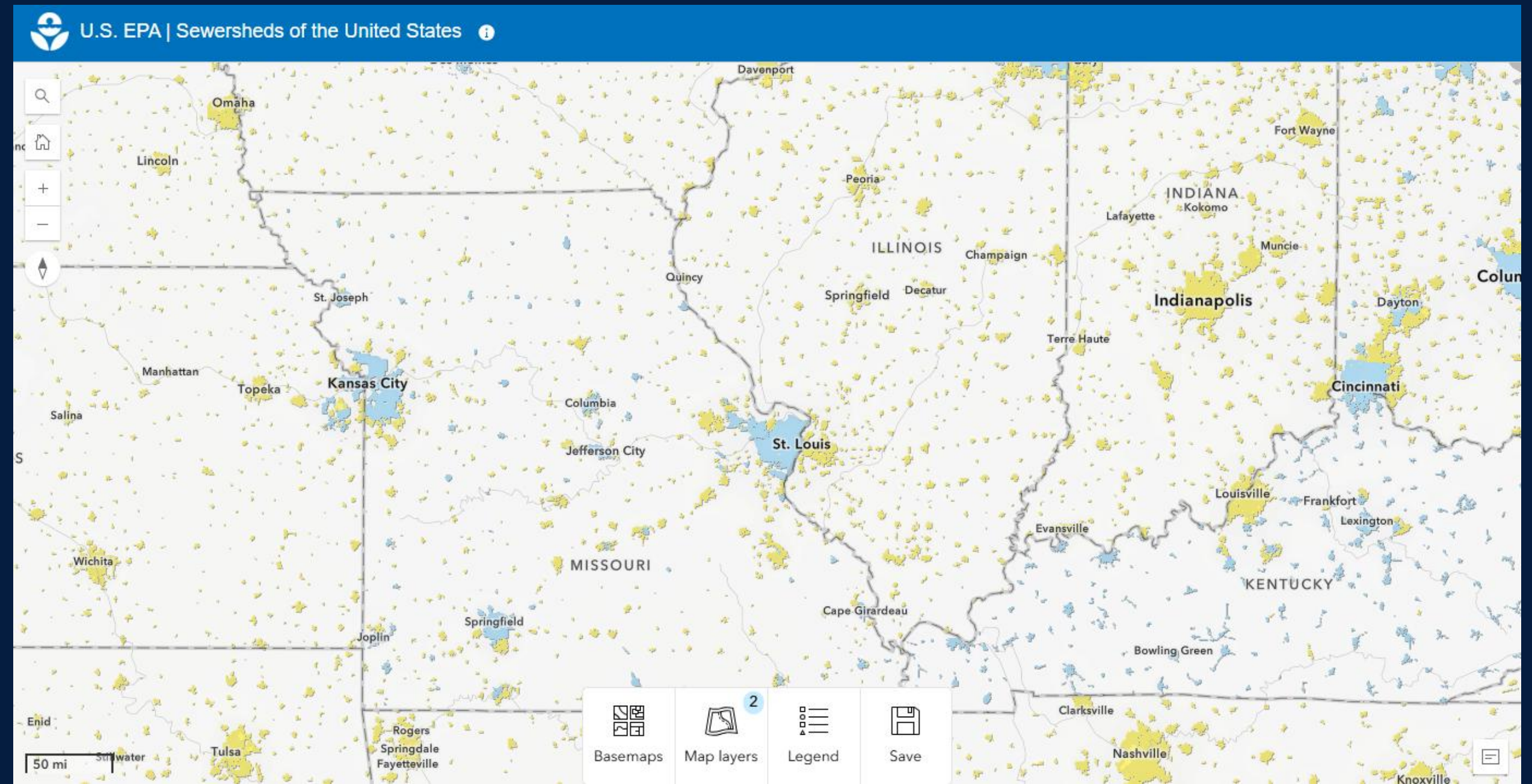
Methodology - Sewershed Creation

- The model assigns a probability to each hexagon
- Hexagons are ranked and assigned to a treatment plant until the aggregate population of the hexagons reaches the target population (reported residential population served)
- This process is done in a loop (one treatment plant after another) starting with plants that serve the smallest populations



Web Map

- Search for Sewersheds
- Inspect Model Results
- Review Utility Sourced Data



Feedback Portal

Publicly Accessible

- Upload
- Edit
- Correct Locations
- Search Locations

An official website of the United States government [Here's how you know](#)

EPA United States Environmental Protection Agency

Search EPA.gov

Environmental Topics Laws & Regulations Report a Violation About EPA

CONTACT US

Submit Edits to EPA's Sewershed Dataset

Guide Upload a Sewershed Edit a Sewershed Address Lookup Move an Endpoint

Select the Sewershed to Upload Data For

Select Sewershed

TRI CITY WPCP (41000001002)

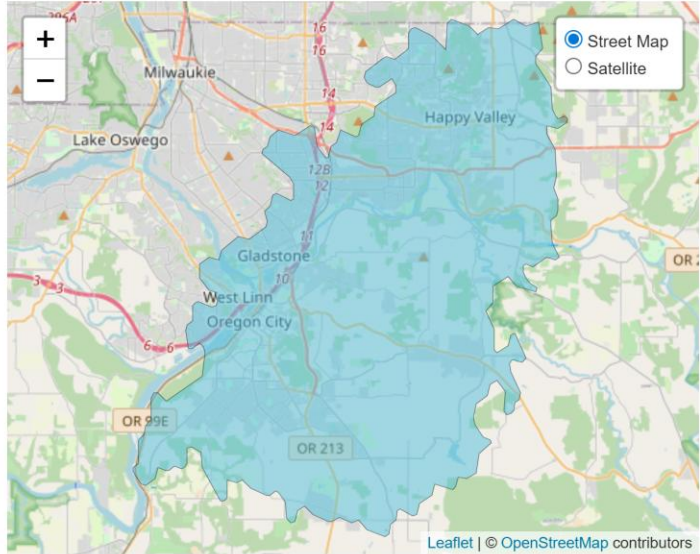
Zipped Shapefile

Browse... No file selected

Load sewershed

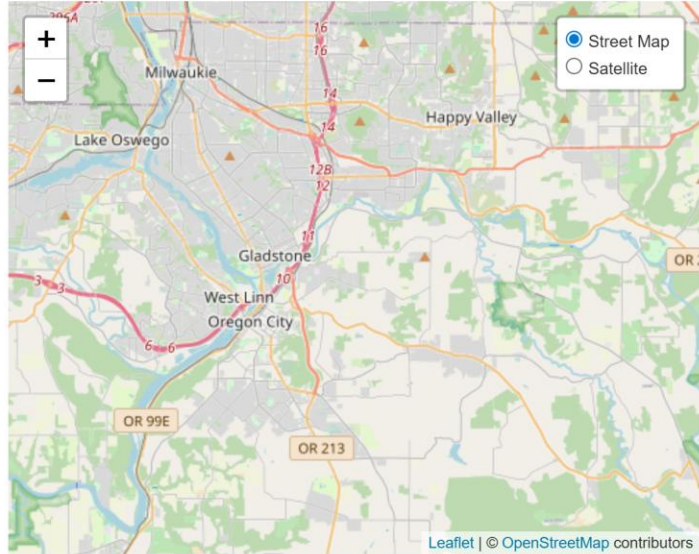
Help

Existing Boundary



Leaflet | © OpenStreetMap contributors

Uploaded Boundary



Leaflet | © OpenStreetMap contributors

Submission Information

Your name:

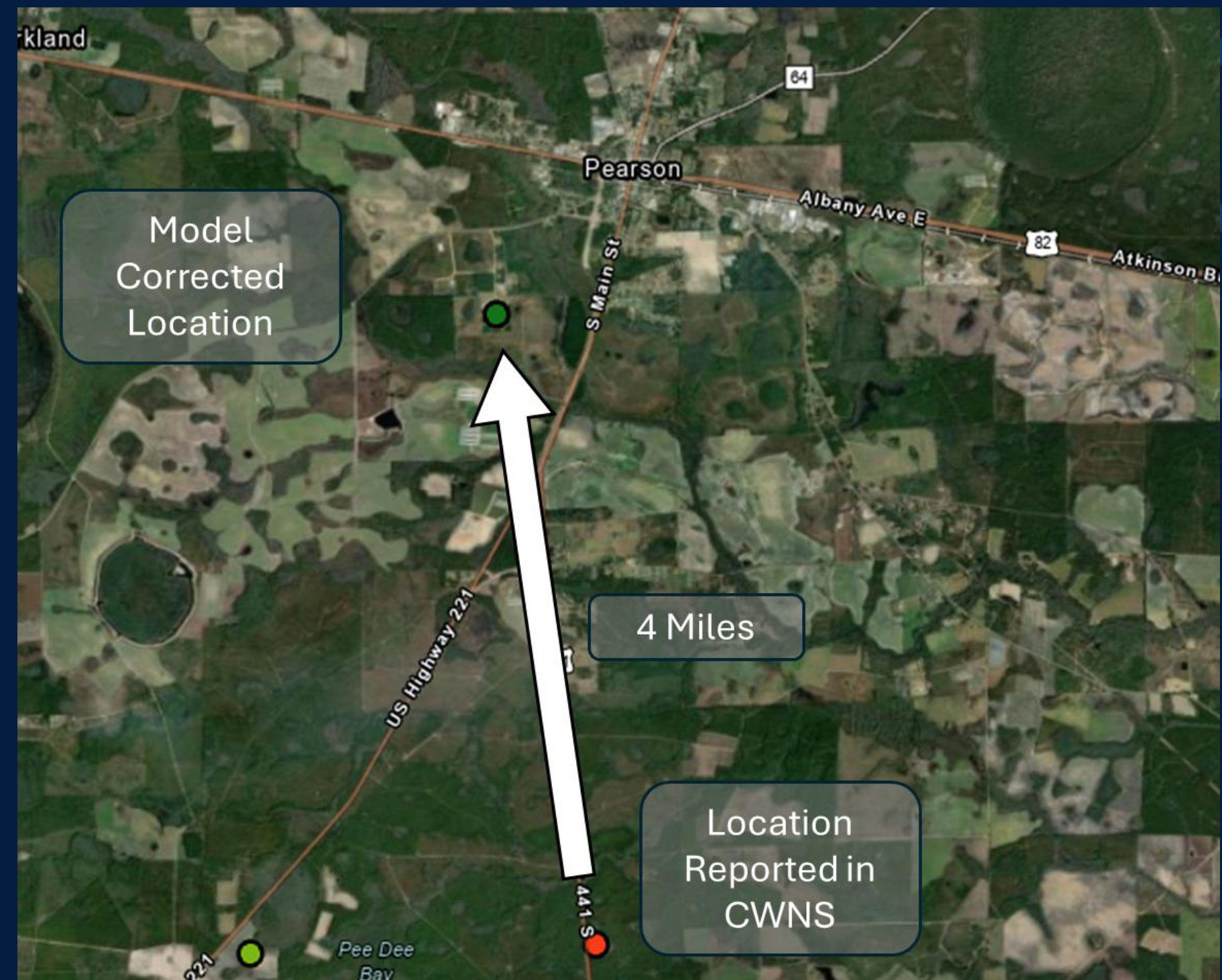
Your email address:

- rconnect-public.epa.gov/Sewershed_Feedback/

Ongoing Development

Model in development to correct geolocations of facilities

- OpenStreetMap
- Parcel Data
 - Zoning
 - Ownership
 - Use Descriptions
- Geographic Names
 - County Subdivisions
 - Places
 - Counties
- Aerial Imagery



Demonstration

- Sewersheds Web Page – <https://www.epa.gov/cwns/sewersheds>
- Web Map – [Link](#)
- Feedback Application – https://rconnect-public.epa.gov/Sewershed_Feedback/

Learn More

- Methods Documentation
<https://epa.maps.arcgis.com/home/item.html?id=c28905ff93dc4f2e9d47d7daf4047d05>
- Data Dictionary
<https://epa.maps.arcgis.com/home/item.html?id=148a867319184160ba0a899bf2f830a5>
- Github Repository –
<https://github.com/USEPA/Sewersheds>



File	Commit	Time
.gitattributes	Current Release	9 minutes ago
.gitignore	Current Release	9 minutes ago
Current_Release.zip	Current Release	9 minutes ago
Sewersheds.Rproj	initial commit	2 years ago
codeinventory.json	Initial commit	2 years ago
contributing.md	Initial commit	2 years ago
license.md	Initial commit	2 years ago
readme.md	readme update	1 minute ago

README

Contributing

MIT license

Data

Current Release: Version 1.0

The current release of the sewersheds data is located in this folder (Current_Release.zip) and will always be maintained using the permanent link: https://github.com/USEPA/Sewersheds/blob/main/Current_Release.zip

As versions are replaced, they will be maintained in the folder 'Version_History'.

Wastewater Infrastructure: Mapping the Nation's Sewersheds

"The [Clean Watershed Needs Survey\(CWNS\)](#) provides an assessment of the capital investments necessary for states, the District of Columbia, and U.S. Territories to meet the Clean Water Act's (CWA) water quality goals over the

QUESTIONS?

CWNS@epa.gov

Email us to be added to the distribution list
for update releases.